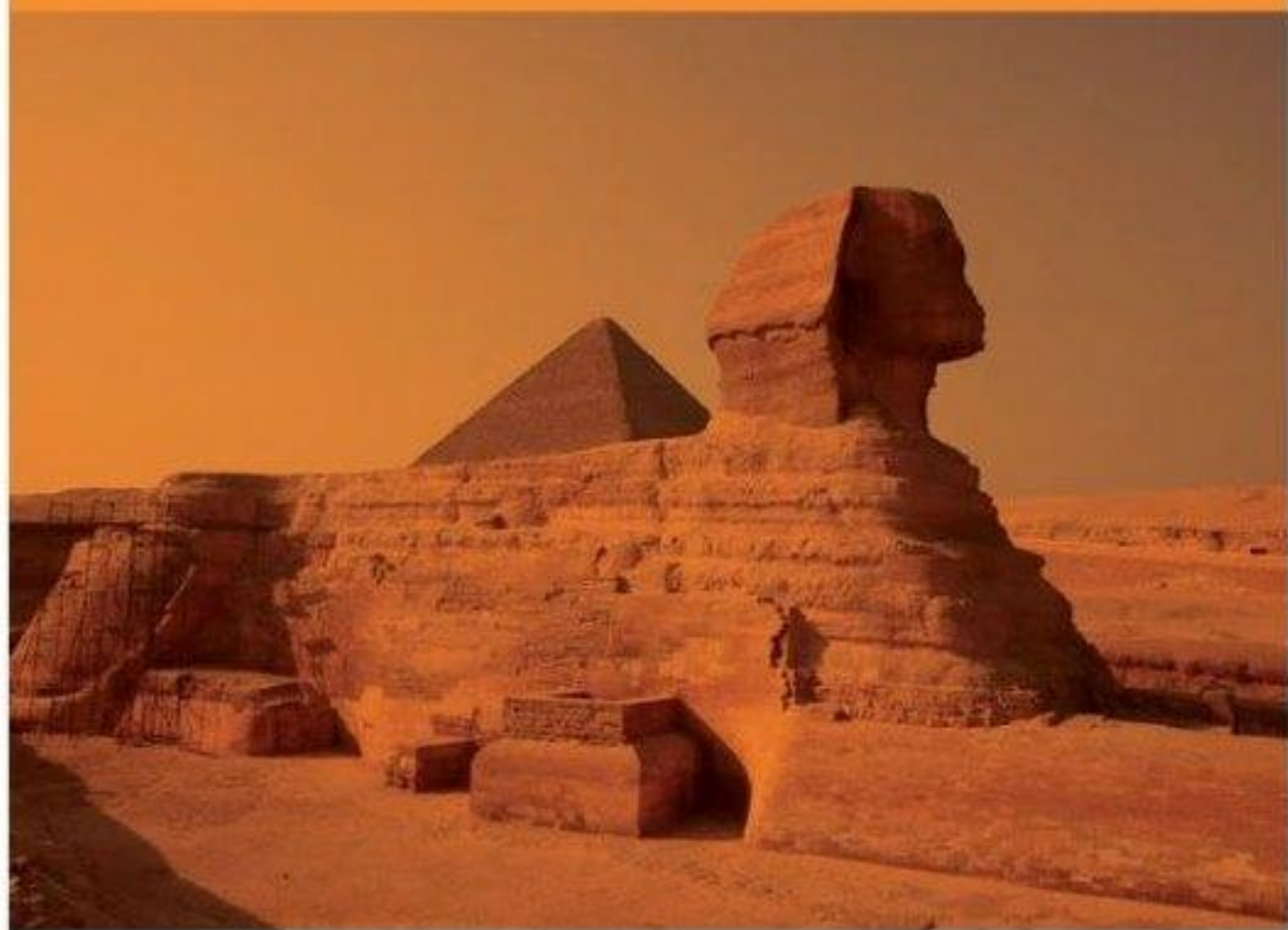


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OPTIMIZATION OF THE CHEMICAL COMPOSITION AND CONCENTRATION OF SOIL SOLUTION OF SOILS OF DESERT ZONE FOR NUTRITION OF PLANTS

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Annotation

The chemical composition and concentration of the soil solution of the soils of the desert zone under intensive farming were studied. As a result of studies, it was found that at a low concentration of soil solution the plant develops poorly, it cannot absorb a sufficient amount of nutrients and, conversely, a high concentration of the solution also negatively affects the plant. The optimal concentration is the most favorable environment for plant development, which helps to accelerate the absorption of nutrients by plants and ensure high crop yields.

The composition and reaction of the soil solution in the practice of agriculture is regulated by the application of fertilizers, soil treatment, and land reclamation. For example, fertilizers are added to enrich the soil with nutrients, soil cleaning is carried out to eliminate salinization and others.

The composition of the soil solution, the concentration and ratios of various compounds in it are subject to seasonal variation during the growing season. This is facilitated by the process of plant nutrition. Especially, in the middle of the growing season (July, August), noticeable changes occur in the composition of the soil solution.

The results obtained are of great theoretical and practical importance in the knowledge of metabolic reactions in the soil-absorbing complex between the absorbed ions and ions in the solution; actions of fertilizers, irrigation and processing technologies on this process; scientifically sound application of fertilizers, as well as in the educational process of higher educational institutions.

Key words: soil, soil solution, anions, cations, concentration, agrotechnical measures, mineral fertilizers, agricultural background, soil-absorbing complex.

Introduction

Relevance of the topic. Improving soil fertility and crop yields is closely associated with the use of fertilizers. However, the use of fertilizers without taking into account the soil properties, especially the composition of the soil solution, will not lead to the desired results, but on the contrary, changes in the composition and pollution of the soil are observed.

Soil solution can be defined as the liquid phase of soils, including soil water containing dissolved gases, organomineral and organic compounds, gases and the finest colloidal sols. Soil solutions are one of the most important categories of natural waters, "the main substrate of life", "the main element of the biosphere mechanism." The significance of studies of a soil solution under pressure from natural and anthropogenic factors is that the soil solution can be optimized for plant nutrition.

Currently, one of the main tasks of agriculture is to maintain the state of the soil solution for optimal plant nutrition. In this regard, the study of the composition and concentration of soil solution, the creation of its favorable condition for plant nutrition is an urgent task of agriculture.

Object and subjects of research. The object of research is the experimental site of the Department of Soil Science of the National University of Uzbekistan named after Mirzo Ulugbek in the desert zone. Subjects of cotton research, newly irrigated gray-meadow meadow soil, agricultural measures, mineral fertilizers, soil solution.

Purpose, research objectives and major versions. The main goal of the research is to develop a method for optimizing the soil solution of the irrigated soils of the desert zone for plant nutrition based on the study of its composition, concentration, osmotic pressure and balance. To achieve this goal, the following tasks were completed: conducting research on newly

irrigated gray-earth meadow soil, determining the chemical composition, concentration, osmotic pressure and equilibrium of the soil solution, detecting changes in the chemical composition and concentration of the soil solution depending on the period of the year.

In natural soils, the soil solution is influenced by the solid, gas phases and natural vegetation cover. Whereas in irrigated soils, natural factors affecting the soil solution also include agricultural techniques (tillage, fertilizer use, irrigation, etc.).

In the study of soil solution should not be limited to the study of only the solid phase of the soil. It should be replenished with scientific materials about the liquid and gaseous phases of soils. Because all three soil phases are constantly interconnected. Especially from the point of view of agrochemistry and plant nutrition, all, without exception, the properties of the soil solution, such as concentration, chemical composition, osmotic pressure and their changes, must know. The vital activity of plants and microorganisms is also impossible without a soil solution, which performs for them both a protective and regulatory function and is a source of nutrition.

The scientific novelty of the topic. For the first time in the conditions of irrigated gray-meadow meadow soil of the desert zone, the composition and concentration of the soil solution were isolated and studied. A change in the chemical composition and concentration of the soil solution was revealed depending on the period of the year.

Literature review

An analysis of literary sources shows that soil solution was studied at the proper level at the Dokuchaev Soil Institute (Russia) P.I. Shavargin (1958-1963), A.A. Kizilova (1955-1967-1970) and at the University of California (United States of America) N.J.Barrow (1972), T.N.Choo (1982). They obtained valuable information on the soil solution of their region.

A number of foreign scientists studied trace elements in soil-pore water (D.M.Bonito, 2005), soil and water chemistry (M.E. Essington, 2004), lysimetric methods for monitoring the conductivity of soil solution and nutrient concentration in tomato greenhouse crops (Francisco Javier Cabrera Corral, Santiago Bonachela Castagno, Maria Dolores Fernandez Fernandez, Maria Rosa Granados Garcia, Juan Carlos Lopez Hernandez, 2016), quantification of pyrophosphate in soil solution by hydrolysis of

pyrophosphatase (Casper Reitzel, Benjamin L. Tour Ner, 2014), the effect of reducing atmospheric precipitation on the chemical composition of forest soils (Arne Verstraten, Johan Neurink, Gerrit Jenou, Natalie Couls, Peter Roskams, Maarten Hens, 2012), chemical properties of the soil solution, rice reaction and water use efficiency with various methods flood management (Jose Bernardo Moraes Borin, Felipe de Campos Carmona, Ibanor Anginoni, Amanda Posselt Martins, Isadora Rodriguez Jaeger, Elio Markolin, Gustavo Cantori Hernandez, Estefania Silva Camargo, 2016), ion leaching and acidification solution in the vadose zone under the soil treated with sewage sludge for agriculture (Ricardo Perobelli Borba, Victor Sanchez Ribeirinho, Otavio Antonio Camargo, Cristiano Albertode Andrade, Carmen Silvia Kira, Alin Rene Cossione, 2018).

According to Smagin (2004), individual states and processes of the interaction of phases and components of soil systems can be described using the categories of equilibrium thermodynamics.

In soil chemistry, there is still an uncertainty in the interpretation of the term "soil solution", which is often understood as meaning different forms of soil moisture in composition and properties. A.E. Vozbutskaya considers the terms "soil liquid phase" and "soil solution" to be analogues.

Explanatory Dictionary of Soil Science (as amended by Rode, 1975) defines a soil solution as "water that is in the soil and contains organic and mineral substances and gases in a dissolved state."

The dictionary "Chemical contamination of soils and their protection" (1991) adds to this that "the soil solution is a homogeneous liquid phase with a variable composition".

R.U. Pearson (cited by Orlov, 1985) calls the soil solution "a quasi-equilibrium solution of electrolytes that exists in the soil under the condition of incomplete saturation with its moisture".

A similar interpretation is also contained in the dictionary "Biosphere: pollution, degradation, protection" (2003): "a soil solution is a solution of chemicals in water that is in quasi-equilibrium with solid and gaseous phases of the soil and partially or completely fills its pore space".

D.S. Orlov (1985) supplements the concept of "soil solution", rightly defining it as the liquid part of the soil in natural conditions. All these definitions reflect some features of such a system as soil solution, but they

cannot be considered strict. For example, it is not clear from them whether all forms of water in the soil can be attributed to soil solution. In particular, are strongly bonded (hygroscopic) and free gravitational moisture related to the soil solution? This question is not idle.

Terminological discrepancies on the one hand and the variety of applied methods for isolating solutions that allow one to obtain different fractions of soil moisture, on the other hand, lead to the fact that researchers in each individual case receive and study systems whose properties are significantly different.

Given that capillary and gravitational moisture are in the same soil, and capillary moisture is one of the sources of formation of the gravitational flow, their composition is generally close. However, there are differences. Since gravitational water is quickly filtered through the soil, not reaching equilibrium with solid phases, its composition less clearly reflects the properties and genetic characteristics of this soil. In addition, it strongly depends on the amount and mode of precipitation and can bear the imprint of the composition of snow, rain or irrigation water, as well as interaction with the ground parts of plants (flushing), litter and an atmosphere that is different in composition from soil air (Snakin, 1989) .

Highly bound moisture, in contrast to gravitational, is stable, motionless, inaccessible to plants and has special physical properties: in particular, it does not dissolve substances soluble in free water. These features make it less informative to study the soil. At the same time, the main part of the liquid phase in most soils is capillary (pore) moisture. This fraction of soil moisture is more stable than the gravitational one, close to the state of quasiequilibrium, is most accessible to plants and is mainly studied in situ measurements.

Therefore, many researchers identify the concept of soil solution with this category of moisture. Other authors consider lysimetric gravity water and capillary soil solution as different fractions of the soil solution that differ in chemical composition and (or) the ratio of components (Samoilova, Demkin, 1976). Apparently, the latter point of view is most correct and all forms of soil moisture present in the soil in situ under natural conditions should be assigned to the soil solution.

The soil solution is not an absolutely homogeneous system, since intensive (independent of the amount of soil) indicators of its properties change in the macroscopic soil space from point to point. This spatial heterogeneity is due to the close relationship between the soil solution and the surface of highly dispersed solid soil particles.

The soil itself is a heterogeneous, porous colloidal capillary system (Smagin, 2003), in which the distribution of ions is regulated by interfacial adsorption interactions that create a concentration gradient of substances near a solid surface, and the retention force of the solution decreases with distance.

Soil moisture is thus represented by several forms of varying degrees of mobility, and its distribution in the soil is directly related to the nature of the pore space. A solution of small pores, not free water, film moisture (according to the terminology of various authors) has specific properties and composition. Anomalous features of non-free, bound water have been established for such characteristics as density, viscosity, dielectric constant, thermal conductivity, dissolving ability.

It is established that in the capillaries of the nanoscale range, water does not behave as prescribed by the classical laws of hydrodynamics, but manifests itself almost like a solid. Flowing, or rather, "squeezing" through a channel with a diameter of 2 nm, the water behaves as a viscous substance and is organized in a series of layers. When compressed, it forms a multilayer structure close to the morphology of liquid crystals. There is an assumption that the water in nanotubes is in the state of aggregation, which differs from liquid and gaseous. Under these conditions, the average number of hydrogen bonds connecting the water molecule with the neighboring ones (coordination number) decreases from 3,8 to 1,86, as a result of which the molecules remain mobile and the water does not freeze even at a temperature of 8 degrees Kelvin, i.e. close to absolute zero .

A.Szent-György (cited by: Kulsky, Dal, Lenchina, 1982) believes that structurally ordered layers of water appear in narrow capillaries near a solid surface.

Structuring extends deep into the liquid phase to a layer thickness of the order of tens and hundreds of molecules (previously it was assumed that ordering is limited only to a monomolecular layer of water adjacent to the surface). The features of water structuring in capillary systems allow us to

speaking with some reason about the capillary state of water. Under natural conditions, it can be observed in pore water, in particular soil solution. One of the reasons for the structuring of water in capillaries is its extremely high surface tension of 0,073 N/m (at 20°C).

Of all liquids, only mercury has a higher surface tension. The lower freezing temperature of pore water is associated with its strong intermolecular contact with the surface of solids, and when it freezes, the properties of those minerals and rocks with which it is in direct contact also change. The solvent capacity of the film of moisture, apparently, differs from the values characteristic of free water.

The concepts of “non-solvent volume” and “negative adsorption” of ions, developed in the works of A.V. Trofimov, were usually associated with film water in the soil. It is based on the idea that water near the surface of the solid phase is directly affected by the surface charge and adsorption forces, as a result of which its properties differ significantly from ordinary ones. In particular, it was experimentally established that within this water film the concentration of anions and some other substances (sucrose) is lower than in the rest of the soil moisture.

At the same time, at a certain distance from the colloidal particle - within the double electric layer - the concentration of dissolved substances is increased relative to the volume of the solution (Pisarenko, Pospelova, Yakovlev, 1964). For example, a decrease in pH was established with a decrease in moisture content in the soil to certain values.

Numerous data indicate that the liquid phase of the soil is not homogeneous, and this circumstance explains the differences in the chemical composition of soil solutions obtained with the application of pressure of various sizes.

Moisture obtained from the soil by centrifugation, which means that it is more firmly bound by the soil (osmotic, sorption, capillary forces), has higher concentrations of most of the studied chemical components compared to solutions extracted by gravity lysimeters.

Similar data are cited by many researchers. For example, studies by Marques et al. (1996) found that soil solutions obtained by vacuum lysimeters ($p = 600 \text{ kPa}$) contain more Si , NH_4^+ , NO_3^- , Cl^- , Mg^{2+} , Al^{3+} compared to solutions obtained by gravity lysimeters. Differences between different

fractions of soil solutions depend on the granulometric composition of the soil.

De Vries, Reinds, Deelstra (1999) showed that in sandy soils differences in soil solutions obtained by different methods are not significant. At the same time, in soils heavier in mechanical composition, the concentration of solutions increases.

The interaction between solute particles and solvent molecules is a prerequisite for the existence of real solutions. This position underlies the chemical theory of solutions proposed by D.I. Mendeleev (1887), who established that both individual particles of a substance and a solvent, as well as products of their interaction, are present in real solutions.

Natural soil solutions are characterized by ionic strength from a few mmol / l to hundreds of mmol/l (in saline soils). So, in the soil solution of solonchak $I = 20-350$ mmol / l, and in solutions from the litter of peaty-podzolic soil, from 0,7 to 8,7 mmol/l. On average, temperate soil solutions are characterized by an ionic strength of about 10 mmol / l (and higher), while weakly concentrated soil solution with an ionic strength of less than 5 mmol / l (up to 0,2 mmol / l) is most typical for highly weathered soils of the tropics (oxisols) (Harter, Naidu, 2001).

In saline soils, as well as in lower mineral horizons, where the content of anions of weak organic acids is low, electrostatic interactions are dominant, and therefore, the use of activity coefficients may be sufficient to calculate real activities, especially using mathematical models that take into account the association of ions.

However, in most cases, in soil solutions, in addition to electrostatic interactions of ions, numerous chemical reactions occur - competing reactions (complexation, protonation). Under these conditions, one element may be present in several different ionic forms, may be represented by a free ion, or may be part of various complex ions.

Studies of soil solutions have been undertaken since the establishment of experimental soil science and were originally aimed at developing methods for separating solutions from the solid phase of the soil.

The first method of displacing soil solutions with distilled water was proposed by T. Schloesing in 1866, and then alcohol was used as a displacer.

Subsequently, a method was developed for squeezing or pressing out soil solutions at high pressures and its modification, the centrifugation method.

Significant distribution in the study of the processes of migration of substances in undisturbed soils received the lysimetric method. Despite the fact that only approximately one can judge soil solutions by lysimetric waters, the method was important for obtaining primary information on the composition and properties of natural waters of various soils and modern processes of migration and accumulation of soil formation products in them.

Historically, the water extract method, which was used by researchers and which is currently the standard method of soil analysis, is among the first.

Of the available studies, the most significant contribution to the experimental study of the interaction of polyvalent cations with fulvic acids was made by M.Schnitzer et al. However, despite the importance and depth of research, they were not oriented toward practical use in calculations, therefore, when setting tasks, the results in the form of conditional stability constants of complexes cannot be directly used as input thermodynamic data.

Given the substantial content and importance of fulvate metal complexes in soil solutions, we dwell in more detail on the method of their experimental study and methods for processing analytical information that provide the necessary thermodynamic constants.

There are very few direct experimental studies of the interaction of fulvic acids with metals in which equilibrium concentrations of fulvic acids and metal ions are represented. Among them is the work of D.S.Gamble, in which the reactions of the interaction of divalent copper with fulvic acids are studied and the equilibrium constants are calculated.

Experimental studies of the equilibrium of fulvic acids with cations are based on the study of the metal distribution between the cation exchange resin and the solution in the absence and presence of a complexing agent, fulvic acid. For this purpose, the Schubert ion exchange method is used.

Thorough studies of the ratio of salts recovered by aqueous extracts and located in soil solutions were carried out by P.I.Shavrygin. It follows from them that the total amount of water-soluble substances in aqueous extracts is higher than in soil solutions; these differences are the higher, the less the solubility of salts. The differences relate more to calcium salts and to a lesser extent to chloride ion.

The method of water extracts, while remaining the main one for controlling the salt state of soils, at the same time also makes it possible by calculation to obtain data on the concentration of soil solutions characterizing the true conditions for the existence of plants in a given soil (N.G.Minashina, 1970).

Karavanova and Malinina (2007) came to the conclusion that the level of vibrations of different chemical elements (and their compounds) in soil solutions from litter and horizons E₁ of peat-podzolic soils differ significantly. Concentrations of Al, Si, Corg vary slightly.

Geographical zonality affects the ratio of precipitation and evaporation, water balance, the nature of vegetation, weathering intensity, the nature of rocks and the type of migration of substances, which inevitably is reflected in the composition of soil solutions according to G.A.Maksimovich (1955), nine latitudinal zones are distinguished, where certain hydrochemical facies of soil solutions. Facies is a site of soil cover, water solutions of which are characterized by the same hydrochemical properties throughout its length.

A.E.Vozbutskaya believes that the terms “soil liquid phase” and “soil solution” are close to each other.

In a soil dictionary, a soil solution characterizes a kind as soil water, containing dissolved organic, mineral substances and gases in its composition.

Generalizing the above, it can be said that a soil solution is the liquid phase of the soil, which contains mineral, organic, and organomineral substances, as well as dissolved gases in ionic, molecular, colloidal forms.

Research methodology

Studies on the allocation, determination of the composition, concentration of soil solutions and their dynamics are determined by the standard methods in agrochemistry. To isolate and study the soil solutions, the Ischerekov-Komarova method of displacing the soil solution was used, where ethyl alcohol was used as the displacer.

It has been shown that plastic or glass tubes 100-150 cm long with an inner diameter of 4 cm filled with a mixture of the test soil with purified quartz sand are optimal for crowding out: ethanol is added in portions of 10-20 ml every two hours. The emitted soil solution is analyzed for the presence of alcohol organoleptically or using a solution of ammonium hexanitrocrylate (NH₄)₂Cit(NO₃)₆, which gives a pink color in the presence of alcohol.

The Ischerekov-Komarova method turned out to be very convenient to use (including the expeditionary version), which led to its widespread use. We also created a device for displacing the soil solution using this technique (Fig. 1).

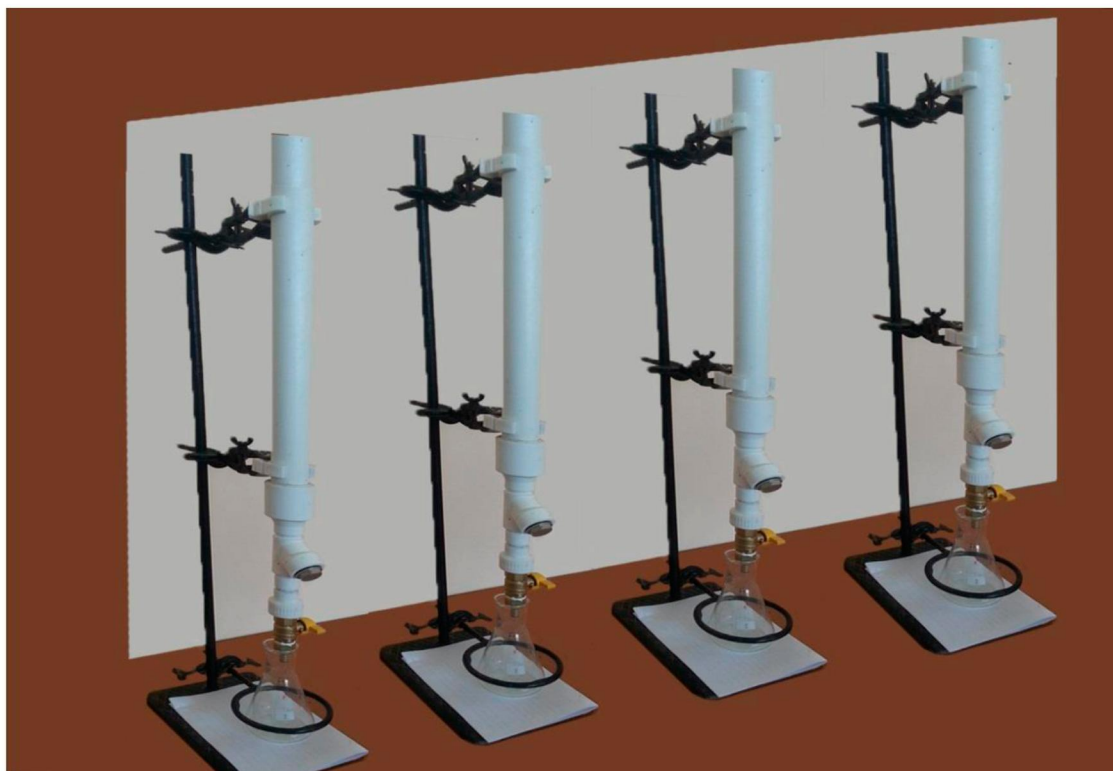


Fig. 1. The device for displacing soil solution

When performing laboratory analyzes, domestic and foreign literary sources were used [1,2,3,4].

Research results

As a result of the analysis of literary sources and our own research, we obtained data on the source, the formation process and the chemical composition of the soil solution of the irrigated soils of the desert zone. According to these data, the salinity of the soil solution ranges from several tens of mg/l to several g/l, depending on the type of soil and geographical location. Below are the results of a study of the chemical composition, concentration, osmotic pressure, and equilibrium of irrigated soils in the desert zone.

Chemical composition, concentration, osmotic pressure and equilibrium of the desert soil solution

The concentration and composition of soil solutions naturally change in the soils of various natural zones. This is due to regular changes in the water regime in the soils of the zonal series.

In semi-desert and desert zones, predominantly saline soils are formed. Their soil solutions are characterized by a high concentration (10-20 g/l). A very high concentration of soil solutions (200-400 g/l) is noted in salt marshes. In the composition of soil solutions of highly saline soils and salt marshes, the main role is played by sodium and magnesium chlorides and sulfates, and the content of chloride ion reaches 100-150 g/l.

Soil solutions of soloncheks, especially soda, have an alkaline reaction, and their concentration changes sharply within the soil profile. In its upper part, the concentration of soil solutions is at the level of 2-9 g/l, in the middle part it increases to 50–60 g/l, and in semi-hydromorphic soloncheks it often reaches 100-120 g/l.

The composition of soil solutions of soloncheks is mainly dominated by chlorides and sulfates of sodium and magnesium. Along with this, they are enriched in colloids and contain carbonates, bicarbonates, aluminates, sodium silicates, dissolved organic matter and ions of iron and aluminum associated with it.

At a low concentration of soil solutions (less than 1 g/l) unfavorable conditions for plants are formed due to a deficiency of biophilic elements. The concentration of the soil solution of more than 5-6 g/l also negatively affects the development of crops, since in the soil there is an accumulation of readily soluble salts - bicarbonates, chlorides, sodium and magnesium sulfates.

Under the influence of salts, inhibition of plants occurs, which can be caused by the following reasons:

- an increase in the osmotic pressure of the soil solution over critical values;
- toxic effects of individual ions on plants;
- violation of plant nutrition conditions.

The osmotic effect as a consequence of increasing the concentration of soil solution is the main factor in the inhibition of plants on saline soils. If the sucking strength of the roots is equal to or less than the osmotic pressure of the soil solution, then the plants cannot absorb soil moisture with nutrients

dissolved in it and die. In addition, temperature and light during salinization can act negatively.

The composition and reaction of the soil solution in the practice of agriculture is regulated by the application of fertilizers, soil treatment, and land reclamation.

The composition of soil solutions depends on precipitation, on the composition of the solid phase, on the quantity and qualitative composition of living and dead plant material in the ground and underground tiers of the biogeocenosis, and on the vital activity of the mesofauna and microorganisms.

Being in close contact with the solid phase of the soil, interacting with precipitation, living organisms and vegetation, the soil solution reflects in its composition all the typical features of this biogeocenosis.

Based on the results obtained, a significant increase in the concentration of Na^+ , Mg^{2+} , Cl^- , SO_4^{2-} is noted in saline soils of the desert zone. The amount of cations in the salt soil solution depends on the solubility of the solid part of the soil and the content of absorbed ions. As a result of biological processes, HCO_3^- , NO_2^- , NO_3^- , H_2PO_4^- , and HPO_4^- ions enter the soil solution.

The chemical composition of the soil solution. In the soil solution of newly irrigated gray-meadow meadow soils of the desert zone. the compounds are predominantly ionized. The Na^+ , Mg^{2+} , Cl^- , SO_4^{2-} ions prevail, and the K^+ , NH_4^+ , and H^+ ions are always present.

The amount of phosphate, chlorine and sulfate ions depends on the dissolution of minerals and the decomposition of plant residues. In saline soils, the levels of sodium and magnesium, chlorides and sulfates are sharply increased.

The concentration of soil solution. The optimal concentration is the most favorable environment for plant growth, allowing the plant to receive higher yields, nutrients and water, as well as grow crops faster. Studies have shown that salinization in recently irrigated pasture soils increases the concentration of soil solution and adversely affects the plant. Under these conditions, the plant is poorly developed and cannot absorb a sufficient amount of nutrients.

Salt causes the death of the plant for the following reasons:

- the osmotic pressure of the soil solution exceeds a critical value;
- toxic effects of certain ions on the plant;
- malnutrition of plants.

According to the results of the study on newly irrigated gray-earth meadow soils of the desert zone, the salinity in the upper layers of some soils (section 1) is insignificant, salts do not exceed 0.140-0.117% (Table 1). However, salts at depths of 70-100 cm are in the range of 1,00-1,40%. The solution of such soils is mainly sulfate, and calcium ion prevails in the cations.

In moderately saline soils (section 2), salts are found in the upper or subsoil horizons. The concentration of the soil solution varies within the profile of moderately saline soils from 1,324% to 1,510%.

Moderately saline soils contain a relatively high level of sulfate ions. The remaining ions in the solution are higher than in slightly saline soils, where the chloride ion is from 0,052 to 0,073%, sodium ion is 0,009-0,054%, magnesium is 0,008-0,032% and calcium ion is 0,018-0,031%.

Table 1: The concentration of the soil solution of newly irrigated gray-meadow meadow soils of the desert zone, %

Depth of horizon, cm	Dense residue	HCO ₃ ⁻	CL ⁻	SO ₄ ⁻²	Na ⁺	Mg ⁺²	Ca ⁺²
Section 1							
0-26	0,140	0,024	0,010	0,043	0,010	0,003	0,017
26-41	0,115	0,025	0,007	0,028	0,006	0,003	0,017
41-71	0,097	0,024	0,006	0,021	0,05	0,002	0,015
71-100	1,105	0,024	0,016	0,054	0,11	0,022	0,021
100-148	0,987	0,016	0,015	0,087	0,008	0,017	0,034
Section 2							
0-30	1,324	0,023	0,073	0,676	0,014	0,008	0,021
30-40	0,770	0,022	0,033	0,353	0,009	0,008	0,020
40-85	0,801	0,021	0,053	0,394	0,054	0,005	0,018
85-110	1,002	0,018	0,073	0,568	0,021	0,032	0,031
110-157	1,510	0,015	0,052	0,725	0,018	0,019	0,039

Studies show that plants absorb nutrients only in the range of 0,2-1,0 mg /l. Further concentration increases the access of nutrients to the plant.

The root system of the plant absorbs nutrients from a highly diluted solution (0,01-0,05%). Under natural conditions, a number of factors, such as saline soils, are affected by 0,02-0,2%.

In the area where the soil sections were laid, cotton was sown from agricultural crops and its condition was studied. According to the results obtained on saline soils, the proportion of ions in the solution is important for the nutrition of cotton. Because each plant needs a certain amount of nutrients during the growing season. It is observed that the normal growth and development of cotton depends on the physiological equilibrium of the soil solution. The physiological equilibrium of the solution is such an optimal state of nutrients in the solution, in which the plant absorbs nutrients from the soil solution to the maximum.

Phenological observations have shown that there is a correlation between the growth and development of cotton and the concentration of soil solution. It is interesting to note that relatively good cotton growth was observed in a soil solution with 0,2% salts. In our opinion, the presence of 0,2% salts in the soil solution increases the solubility of phosphates and ensures the physiological balance of the solution. The presence of nitrogen, phosphorus and potassium in the solution determines the rate of assimilation of other nutrients. Increased nitrogen nutrition accelerates the transmission of many plants P, K, Ca, Mg, Si, Fe, Mn and Zn. If the amount of nitrogen increases, the opposite happens.

Elevated levels of phosphorus in solution reduce the penetration of elements such as copper, manganese and iron into the plant. When a plant is well provided with nitrogen, phosphorus and potassium, its need for trace elements increases dramatically.

A concentration of soil solution of more than 5-6 g/l has a negative effect on the development of crops. Under these conditions, sodium and magnesium readily soluble salts of bicarbonates, chlorides and sulfates accumulate in the soil.

Osmotic pressure of the soil solution. The osmotic pressure of the soil solution is very important for plant nutrition. If the osmotic pressure of the soil solution is equal to or greater than the osmotic pressure of the plant cell, the penetration of water will stop.

According to the results obtained, the osmotic pressure depends on the concentration of the soil solution and the degree of dissociation of the dissolved substances. Also, the osmotic pressure of the soil solution is proportionally affected by fertilizer application and salinity of the soil.

A high concentration of soil solution in saline soils increases the osmotic pressure of the solution, making it difficult for the plant to absorb nutrients and water. The plant is especially sensitive to high concentrations at a young age. Many plants cannot withstand the concentration of the solution above a certain limit.

The influence of osmotic pressure associated with an increase in the concentration of soil solution is the main factor in the death of plants. If the root absorption force is equal to or less than the osmotic pressure of the soil solution, the plant will not absorb and retain the moisture of the soil in which the nutrients are dissolved.

The accumulation of salts in desert soils leads to an increase in the osmotic pressure of the soil solution, as a result of which a physiological dry state of the soil is observed. Under these conditions, the plant cannot absorb the water it needs, even if there is enough moisture in the soil. At the same time, under salinization conditions, temperature and light negatively affect the osmotic pressure of the soil solution.

In non-saline soils, the osmotic pressure ranges from 10,32-10,65 atm in spring and 13,41-13,73 atm in July-August. One of the main reasons for this change is soil moisture. It should be noted that normal fluctuations in the soil solution are created only when the osmotic pressure of the soil solution is 2-3 atm.

The reaction of the soil solution in the studied soils varies between neutral, slightly alkaline and alkaline.

The composition and concentration of saline soils in the desert zone can be controlled by fertilizing, tillage and land reclamation.

Conclusions

1. The study was conducted on weak and moderately saline gray-earth meadow soils of the desert zone.

2. In the studied soils of the desert zone, humus is 1,20-1,25%; total nitrogen 0,086-0,088%; phosphorus ranges from 0,10-0,12% and a potassium content in the range of 2,11-2,10%.

3. Mobile forms of nitrogen, phosphorus and potassium are the main sources of nutrition for plants. Nitrate nitrogen in the upper layers of the soil is 27,7-28,1 mg/kg, mobile phosphorus 35,3-35,5 mg/kg, potassium 215,0-220,3 mg/kg, which means these soils are poorly provided mobile nutrients.

4. The chemical composition of the soil solution is directly related to the parent rocks and climatic conditions, the soil solution of the desert zone has a moderate mineralized chemical composition.

5. The total amount of readily soluble salts in the soil solution of soils in the desert zone is more than 0,3%.

6. In irrigated agriculture, a high concentration of soil solution is observed due to increased biological activity in soils.

7. The composition of the soil solution also depends on the nutrition of crops. Significant changes are observed in the soil solution in July-August, when vegetation is exposed. During this period, the soil solution contains an increase in the nutrient content in plants and increases the absorbing function of the root. In subsequent periods of vegetation, calcium ions decrease in the soil solution, and potassium ions increase as a result of direct exposure to the plant. As a result, the ratio of potassium ion to calcium ion increases. The process of penetration of nutrients to the root depends on this ratio: the greater the speed in the solution, the stronger the ability of the root to absorb, or vice versa.

8. To optimize the composition and concentration of newly irrigated gray-meadow meadow soils of the desert zone for plant nutrition, it is recommended to use fertilizers in the norm $N_{200}P_{175}K_{125}$ and irrigation norms, which provide 70-80% of soil moisture.

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